Summary of the baseline Trust reads (TRUST investigators):

N.records = 416

N.patients = 416

Q: How many with disease typical/suggestive of TB, how many with atypical disease?

Abnormal, highly suggestive of TB: 299

Abnormal, may be compatible with TB: 93

Abnormal, not compatible with TB: 18

Normal: 6

Q: How many with cavitary disease?

Yes: 283

No: 122

Unknown: 11

Q: How many with unilateral vs bilateral disease?

Yes, bilateral: 261

Yes, unilateral: 134

No: 21

Q: How many with miliary or disseminated TB?

No: 407

Yes: 5

Unknown: 3

Q: How many with Granuloma?

No: 255

Yes: 159

Unknown: 2

Q: How many with Effusion?

No: 343

Yes: 70

Unknown: 2

Q: How many with Lymph?

No: 256

Unknown: 100

Yes: 60

Sextant Analysis (TBP data):

We noticed that very few patients have complete sextant information - only 19 out of 260. We were wondering what might be the reason behind the missingness? Also, just to add more information, we ended up with 260 patients out of the original 329 because 69 patients did not have any imagingstudy_id.

Ideally, we should have the following values for each patient: 'Upper Right Sextant', 'Upper Left Sextant', 'Middle Right Sextant', 'Middle Left Sextant', 'Lower Right Sextant' and 'Lower Left Sextant'. But only 19 patients have all the 6 values, the rest have 5 or lesser. If you would like to check these on your end, please check the Patient Identifier 'T0066': they have all the 6 values in the sextant column whereas patient 'T0043' only has 3 values in the sextant column.

Response from Urvi (NIH Team):

For each individual X-ray image, we only highlight the sextants that exhibit any abnormalities. As such, not every sextant will necessarily show abnormalities in every patient. Therefore, it is completely normal if an X-ray image report only includes 5 or fewer sextants with noted abnormalities. Additionally, for the 69 patients who do not have images, we have checked the Box folder ‘Chest X-Ray DICOM Files’ where images were transferred in May 2023. This folder does not have images for 67/69 patients.In addition, for the 2 missing patients for whom the images exist in the above Box folder and they are also uploaded to the portal, we can follow up with the radiologists about what they saw with these two images.However, we also found that the images were uploaded as DICOM files within the subfolder titled "DICOM Transfer_7.19.2024" for 41/69 patients from Study 1.As we evaluate the second batch of images received, we will make a note of all the imaging files that belong to Study 1 and ensure they are properly uploaded.

PLI vs Timika Score:

There are records where PLI is not the same as the Timika score. The cases where the PLI and Timika are the same, we have no cavity. When there is cavity, PLI and Timika are different – Timika is 40 points greater.

N.patients for TBP = 329

69 patients had no imagingstudy_id, so we removed them. N.patients = 260

Some patients had duplicate imagingstudy_id and Days since Imaging date, we kept the record where any_cavity was present. 'any_cavity' was calculated by summing smallcavities + mediumcavities + largecavities, if the sum > 0, any_cavity = 1 otherwise any_cavity = 0.

Some patients still had more than 1 imagingstudy_id, in that case we kept the earliest imagingstudy_id.

Note from Ms. Sue: "manual reads have not yet taken place for the follow-up (post-treatment) chest x-ray. I would remove the instances in which 'Days since the imaging date' is higher, thus delete the more recent images"

Observation: There were patients with different Days from Imaging date that had a higher PLI associated with their latest imagingstudy_id.

Unilateral vs Bilateral: (N.patients = 254)

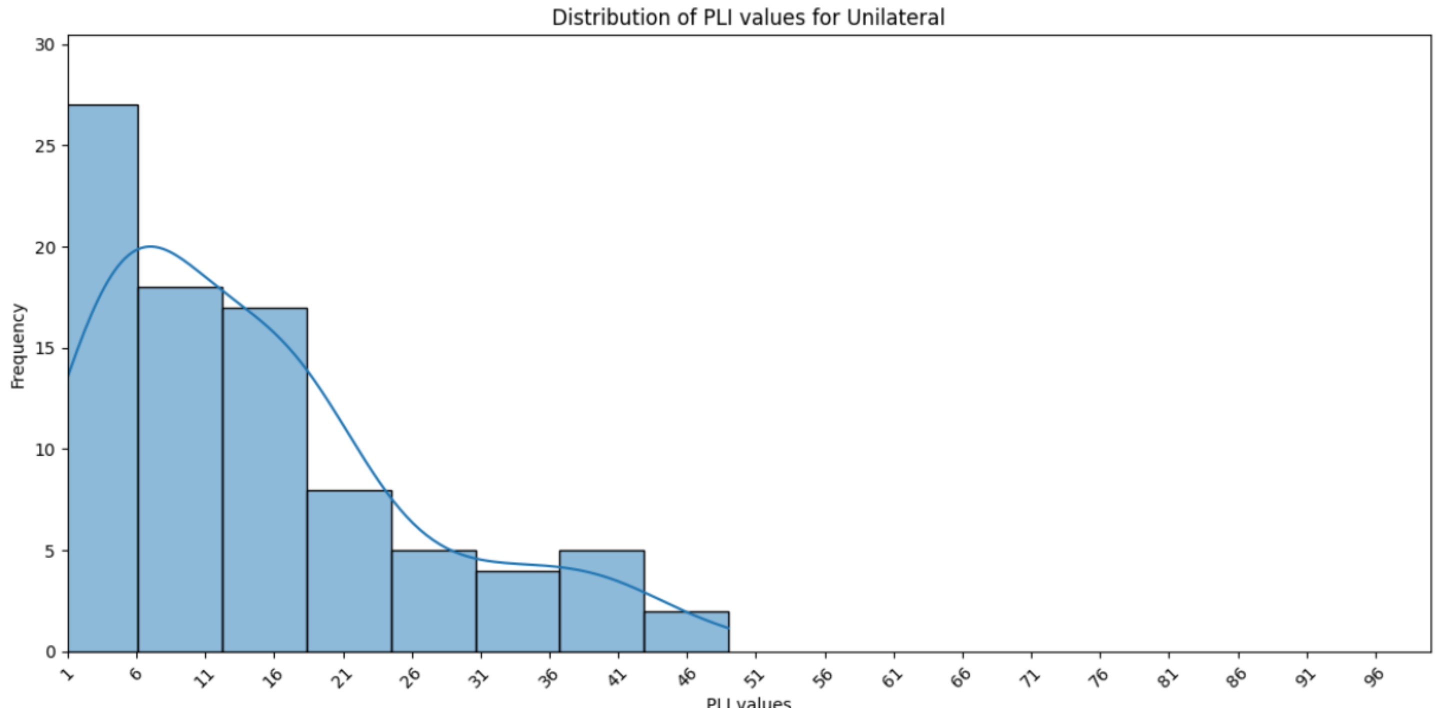
6 patients either have empty value or a NaN (260 – 6 = 254)

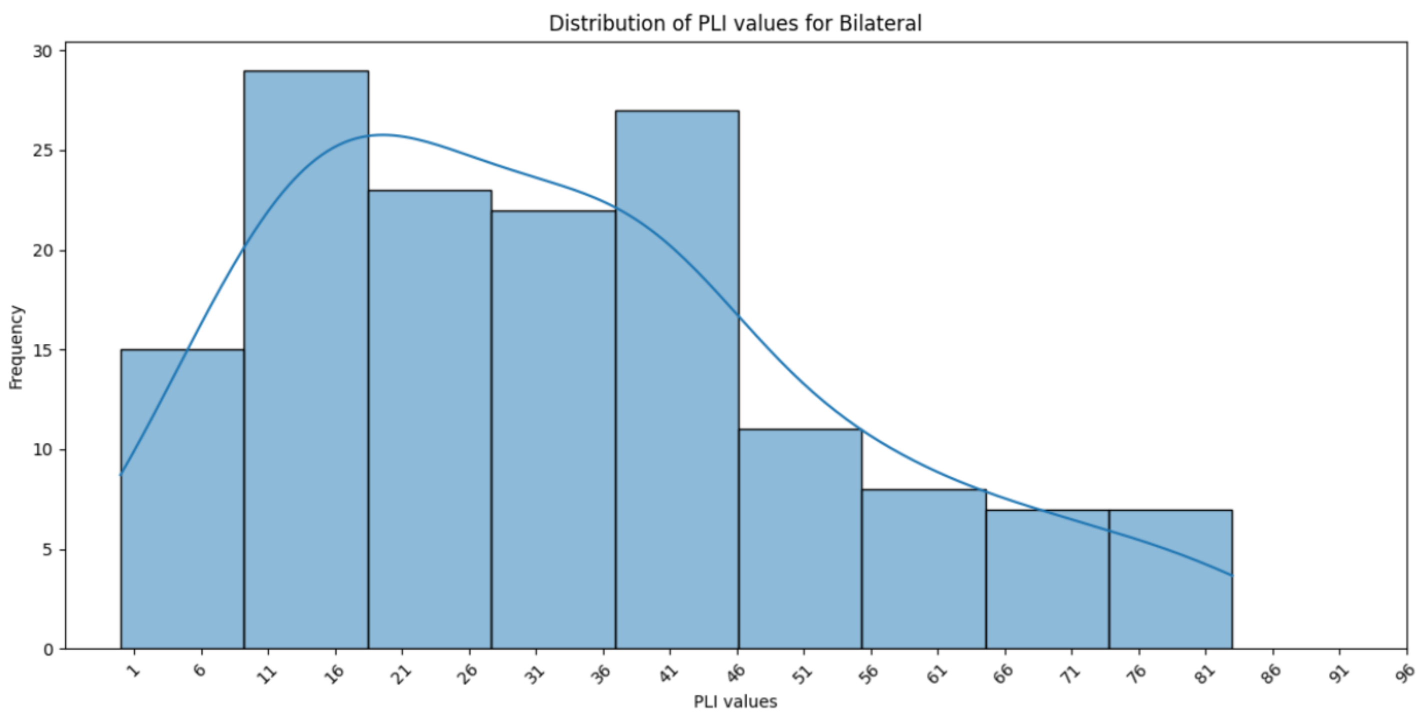
N.patients, none (TRUST labels): 19

N.patients, bilateral (TRUST labels): 149

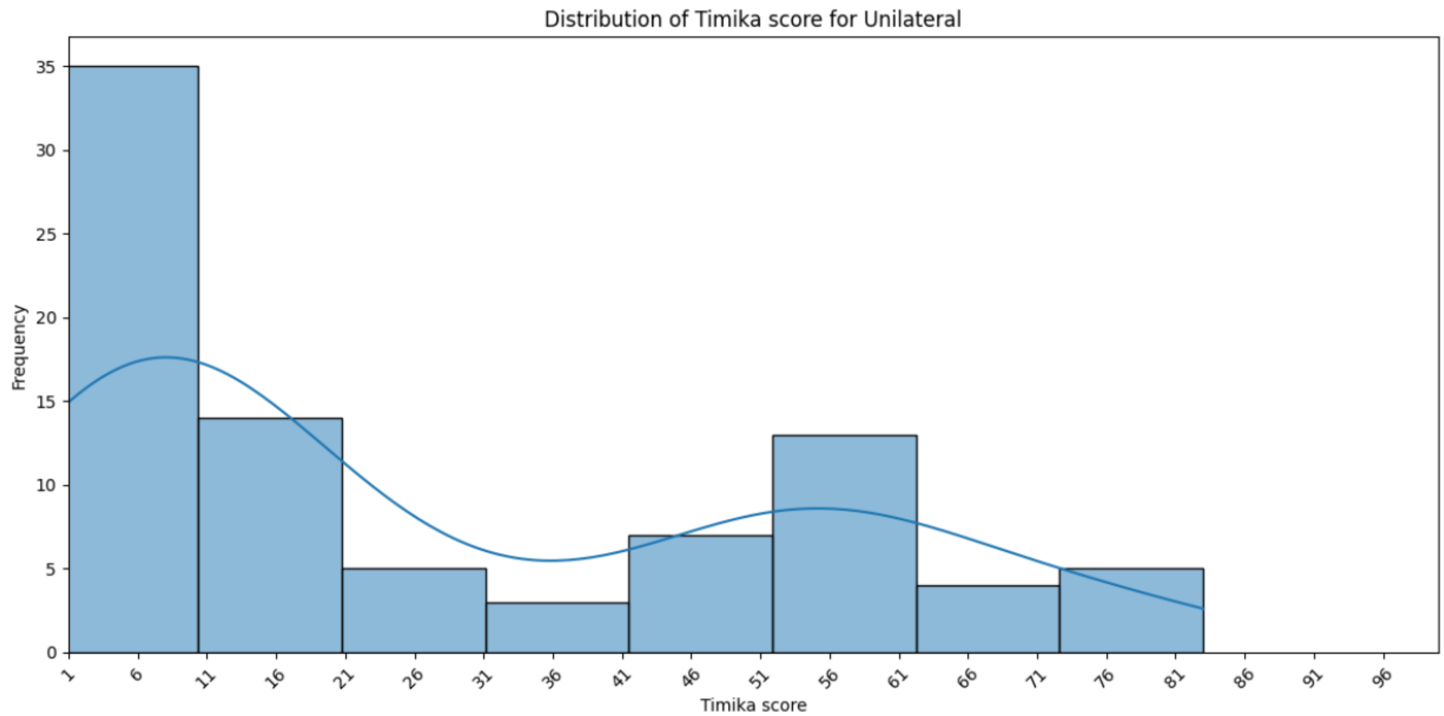
N.patients, unilateral (TRUST labels): 86

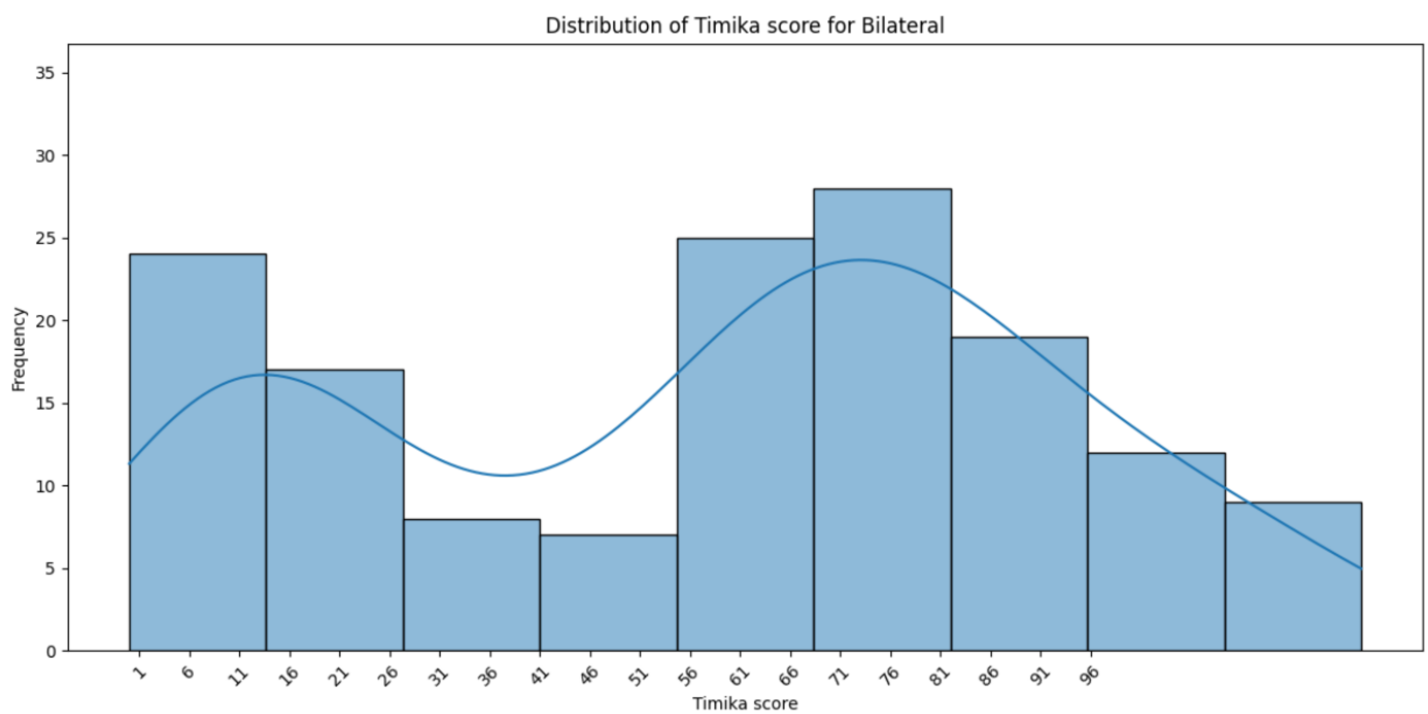
PLI Distribution (Unilateral vs Bilateral):





Timika score Distribution (Unilateral vs Bilateral):





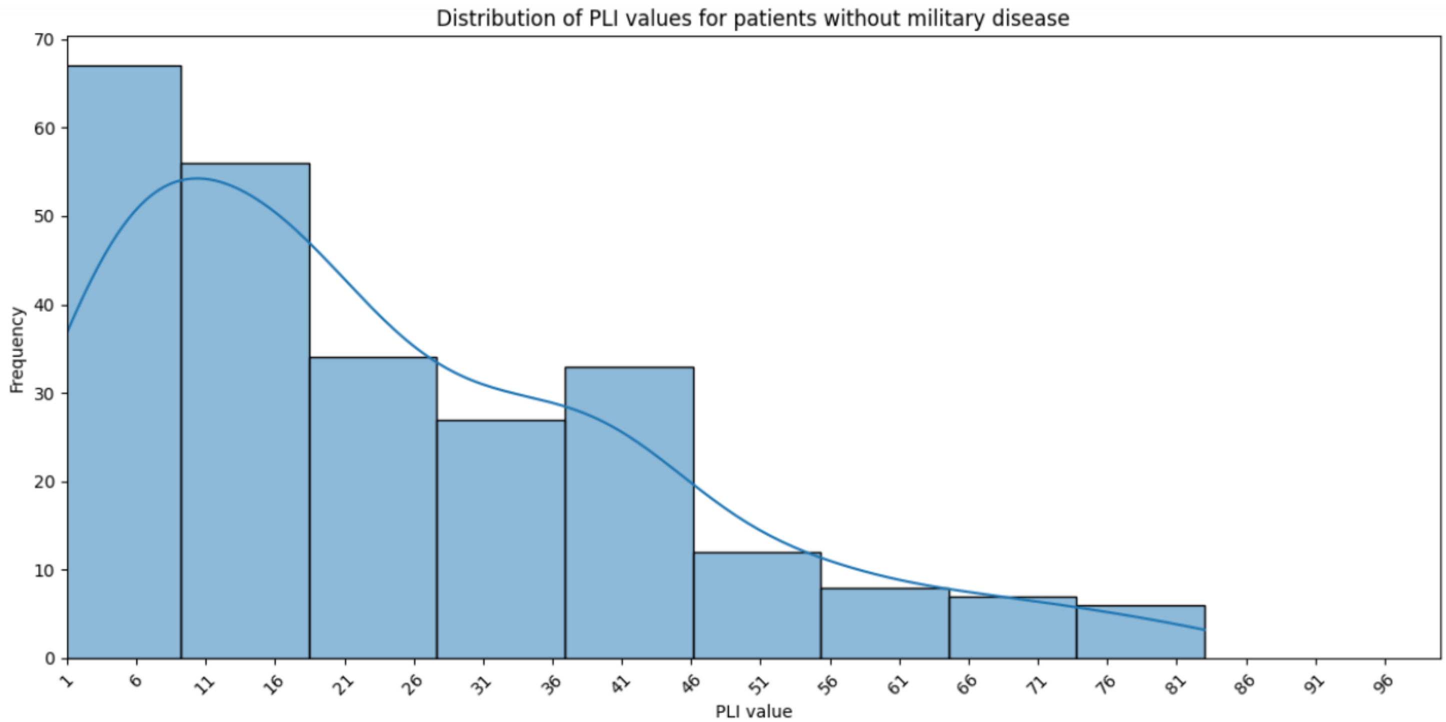
Military Disease:

N.patients, with Military disease (TRUST labels): 2

N.patients, without Military disease (TRUST labels): 250

N.patients, unknown (TRUST labels): 2

PLI values for the patients with Military disease: 46, 79



Presence of Cavity:

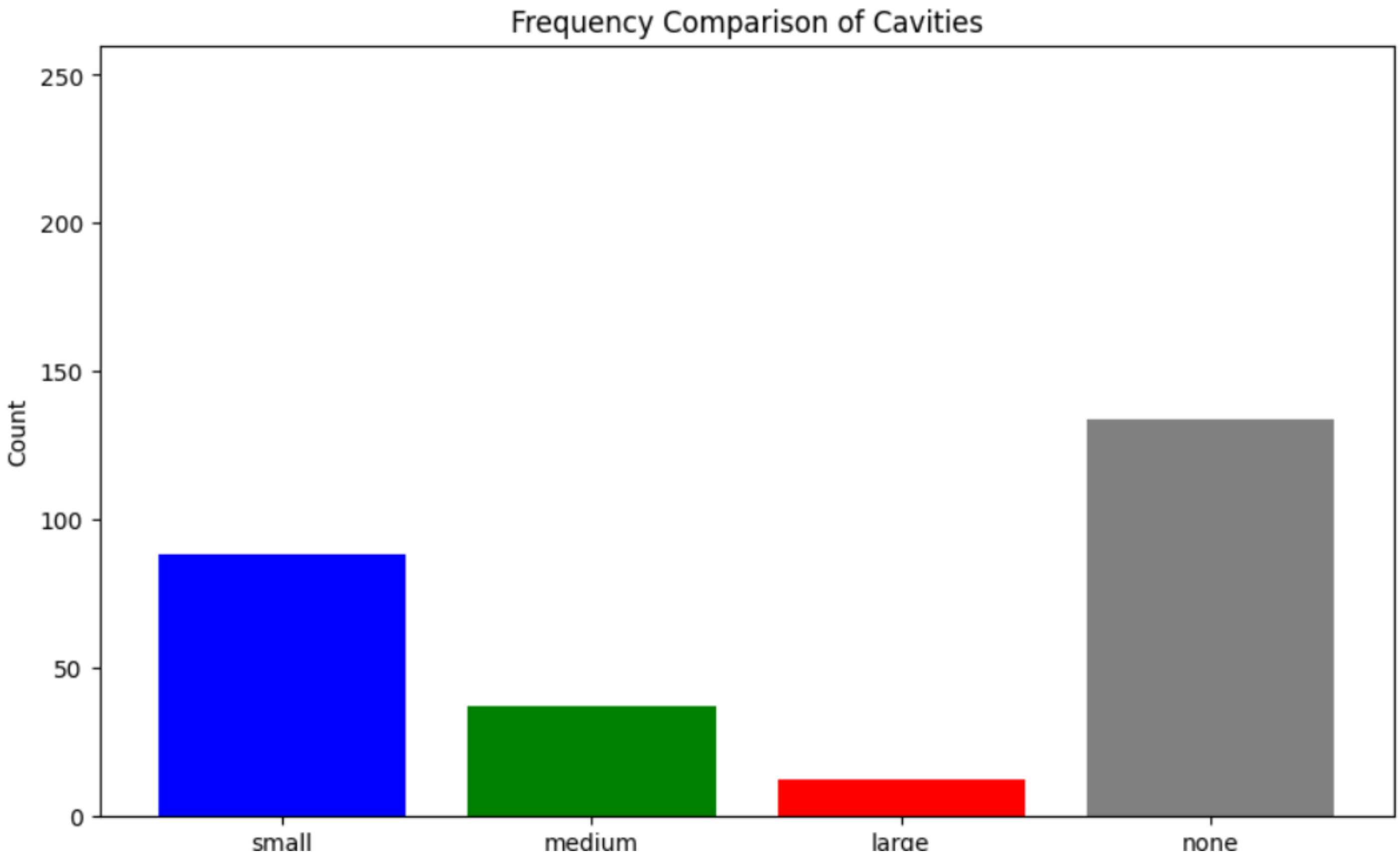
Percentage of matching entries between the TBP reads and the TRUST manual reads: 73.62%

Confusion Matrix:

	trust_manual_reads: 0	trust_manual_reads: 1
tbp_any_cavity: 0	79	52
tbp_any_cavity: 1	15	108

Small vs Medium vs Large cavity frequency comparison:

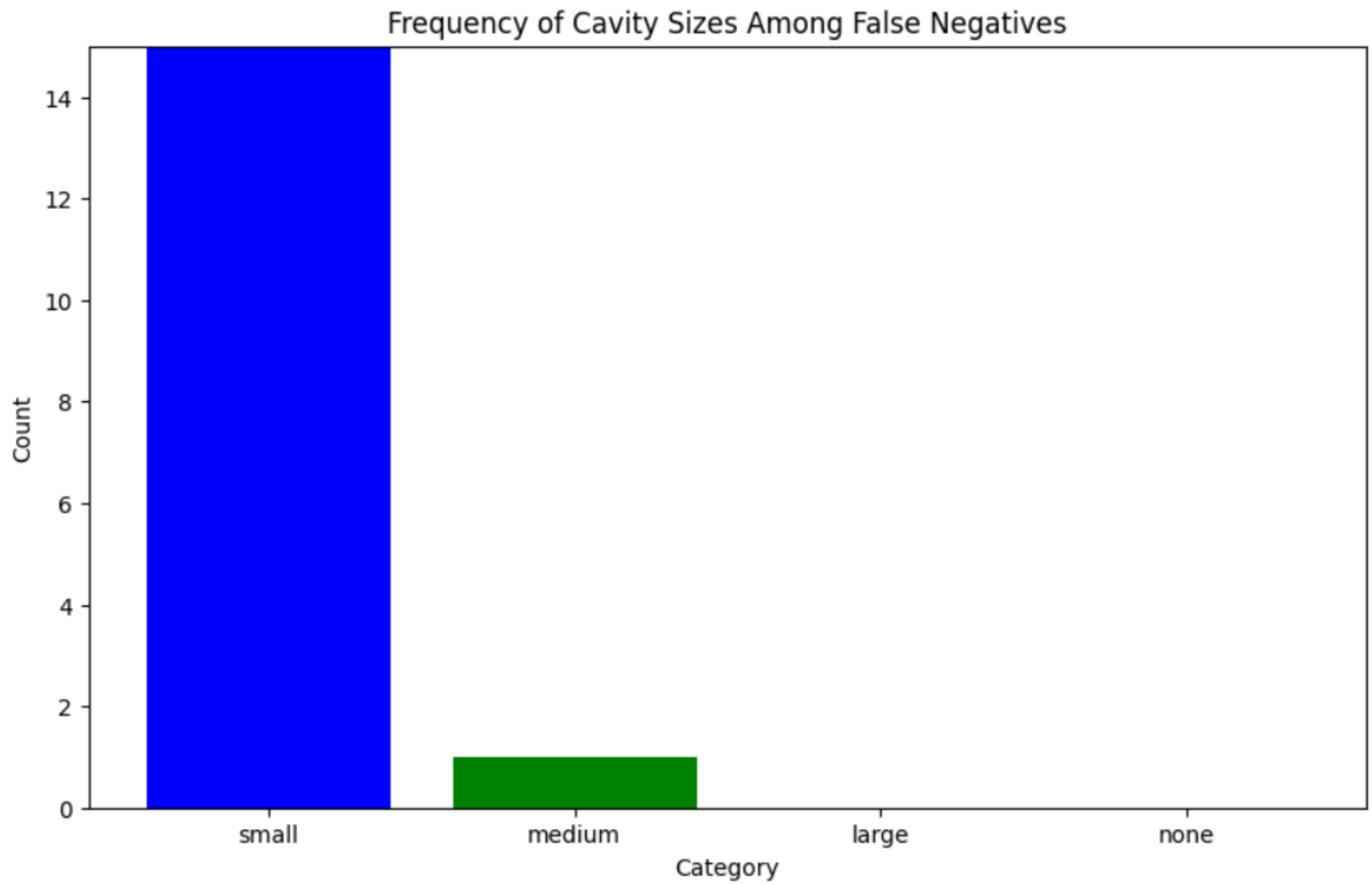
```
   category  count
0     small     88
1    medium     37
2     large     12
3      none    134
```



Frequency comparison in the False Negatives:

Note: 1 patient has both small and medium cavity size recorded at the same time.

Total False Negatives: 15



TRUST CXR Data Summary:

Available DICOMs on OneDrive:

N.images = 1351

N.patients = 454

Note: Approx. 6% images have 'ViewPosition' information in the DICOM tags. We need to filter out the images to keep only the Frontal / PA view. To resolve the missingness in the ViewPosition DICOM tag, we extracted labels (1A, 1B, 2C, etc.) and then manually viewed them to filter out the Frontal / PA CXR views.

Labels extracted from Image name: 1A, 1B, 1C, 1D, 1E, 2A, 2B, 2C, 3A, 3B

Label counts:

1A: 442

1B: 450

1C: 87

1D: 3

1E: 1

2A: 177

2B: 177

2C: 12

3A: 1

3B: 1

Note: Based on discussion with the BMC group, we have filtered out 1B and 2B as Frontal view CXRs.

After filtering the images by the labels 1B and 2B:

N.images = 627

N.patients = 453

DICOM Tags of interest: ViewPosition, PatientOrientation, PhotometricInterpretation, Rows, Columns, PixelSpacing, PixelData, ImagerPixelSpacing, Modality

Overview of DICOM tags available and Missing Data Analysis:

DICOM Tag	Percent Missing
PerformedProcedureStepDescription	100
PatientSize	100

PatientBirthTime	100
OtherPatientIDs	100
OtherPatientNames	100
PatientAge	100
EthnicGroup	100
DeviceSerialNumber	100
PlateID	100
DetectorID	100
RequestingService	100
RequestingPhysician	100
PositionerSecondaryAngle	100
AdditionalPatientHistory	100
PositionerPrimaryAngle	100
PregnancyStatus	100
AcquisitionDeviceProcessingDescription	100
ProtocolName	100
MilitaryRank	100
IrradiationEventUID	100
AdmittingDiagnosesDescription	100
PerformingPhysicianName	100
PerformedProcedureStepStartDate	100
PerformedProcedureStepStartTime	100
PerformedProcedureStepID	100
OperatorsName	100
ContrastBolusAgent	100
InstitutionalDepartmentName	100
AcquisitionDate	100
InstitutionName	100
PatientWeight	100
AccessionNumber	100
ContentTime	100
AcquisitionTime	100
SeriesTime	100
StationName	100
StudyDescription	100
ContentDate	100
InstitutionAddress	100
SeriesDescription	100
PatientComments	100
SeriesDate	100
ImageComments	100
ReferringPhysicianName	95.1
StudyID	95.1
StudyDate	95.1
StudyTime	95.1
PatientName	95.1
PatientBirthDate	95.1
PatientSex	95.1

PatientID	94.7
ConversionType	94.6
Laterality	94.6
DeviationIndex	93.8
PixelAspectRatio	93.8
ImagesInAcquisition	93.8
SpatialLocationsPreserved	93.8
ViewPosition	93.8
CassetteSize	93.8
ReferencedSOPInstanceUID	93.8
MappingResource	93.8
ContextGroupVersion	93.8
ContextIdentifier	93.8
ReferencedSOPClassUID	93.8
DetectorManufacturerName	93.3
DetectorManufacturerModelName	93.3
InstanceCreationTime	93.3
InstanceCreationDate	93.3
ImageLaterality	81.5
PatientOrientation	81.3
ImageDisplayFormat	75.4
Trim	75.4
BorderDensity	75.4
FieldOfViewShape	75.4
FieldOfViewDimensions	75.4
DetectorConfiguration	75.4
FilmOrientation	75.4
ExposureControlMode	75.4
ExposureControlModeDescription	75.4
AnnotationDisplayFormatID	75.4
PresentationIntentType	68.8
PixelIntensityRelationshipSign	68.8
RescaleType	62.5
RescaleSlope	62.5
RescaleIntercept	62.5
ScheduledProcedureStepID	56.3
StudyStatusID	49.2
CodeMeaning	43.1
CodingSchemeDesignator	43.1
ImageAndFluoroscopyAreaDoseProduct	43
XRayTubeCurrentInuA	42.9
ExposureTimeInuS	42.9
ColumnAngulation	42.9
DateOfLastCalibration	42.9
DetectorConditionsNominalFlag	42.9
DateOfLastDetectorCalibration	42.9
Grid	42.9
FieldOfViewOrigin	42.9

FieldOfViewRotation	42.9
FieldOfViewHorizontalFlip	42.9
TimeOfLastDetectorCalibration	42.9
GridFocalDistance	42.9
FilterType	42.9
Exposure	42.9
KVP	42.9
TableMotion	42.9
TableAngle	42.9
TableType	42.9
ExposureTime	42.9
RadiationMode	42.9
XRayTubeCurrent	42.9
ExposureInuAs	42.9
AveragePulseWidth	42.9
RadiationSetting	42.9
DetectorTemperature	37.1
AcquisitionNumber	36.6
ExposureIndex	36.6
ManufacturerModelName	36.6
DetectorType	36.2
RelativeXRayExposure	30
PositionerType	18.3
DetectorDescription	18.3
AcquisitionDeviceProcessingCode	18.3
DistanceSourceToDetector	18.3
CodeValue	13.1
TargetExposureIndex	12.1
PixelSpacing	12.1
Sensitivity	11.6
PixelIntensityRelationship	11.6
PresentationLUTShape	11.6
LargestImagePixelValue	11.5
SmallestImagePixelValue	11.5
Manufacturer	5.4
ImageType	5.4
BodyPartExamined	5.4
SoftwareVersions	5.4
ImagerPixelSpacing	5.4
LossyImageCompression	5.4
Unnamed: 75	5.4
BurnedInAnnotation	5.4
SpecificCharacterSet	4.9
DerivationDescription	0.5
PixelData	0
PixelRepresentation	0
patient_id	0
SOPClassUID	0
SOPInstanceUID	0

Modality	0
StudyInstanceUID	0
SeriesInstanceUID	0
SeriesNumber	0
InstanceNumber	0
SamplesPerPixel	0
PhotometricInterpretation	0
PlanarConfiguration	0
Rows	0
Columns	0
BitsAllocated	0
BitsStored	0
HighBit	0
view	0

Note: Even though the missing analysis shows data available in PHI columns like 'PatientName, PatientID, PatientBirthDate'. Upon viewing the values, it seems that all those are the same and possibly dummy values.

QC the 1B and 2B view DICOMs using TBP QC algorithm to detect Outlier images vs Normal images.

The following steps are used in the algorithm to detect outliers:

1. Normalize the image into a standard size and intensity range.
2. Register the image to a CXR atlas.
3. Transform the image into an approximation with a precomputed principal component basis.
4. Predict if the image is an outlier determined by how well the image fits to the model.

N.Total (1B+2B) = 627

N.Normal = 566

N.Outlier = 61